Davies-Bouldin Index (DBI): Formula and Numeric Example

The **Davies-Bouldin Index (DBI)** is an internal validation metric for evaluating clustering quality. It measures the balance between intra-cluster compactness and inter-cluster separation. A **lower DBI** indicates better clustering (tight clusters that are well-separated). Below, we define the formula and demonstrate it with a numeric example. Key sources are cited where relevant.
(Note: Minor formula variations exist in sources, but we use the consensus standard.)

1. Formula

The standard DBI formula is:

$$DBI = \frac{1}{k}\sum_{i=1}^{k} \max_{j \neq i}\left(\frac{s_i + s_j}{d(c_i, c_j)}\right)$$

Symbols:
k: Number of clusters.
s_i: **Average intra-cluster distance** for cluster i. Calculated as the mean Euclidean distance between each point in cluster i and its centroid:

$$s_i = \frac{1}{|C_i|}\sum_{x \in C_i} \|x - c_i\|_2$$

where $|C_i|$ is the number of points in cluster i, and c_i is the centroid of cluster i.
$d(c_i, c_j)$: **Euclidean distance between centroids** c_i and c_j:

$$d(c_i, c_j) = \|c_i - c_j\|_2$$

$\max_{j \neq i}$: For cluster i, find the cluster j ($j \neq i$) that maximizes the ratio $\frac{s_i + s_j}{d(c_i, c_j)}$. This identifies the "most similar" (least separated) cluster to i.
Key Properties:
Lower DBI is better: Values closer to 0 indicate tight, well-separated clusters. High values suggest overlapping or diffuse clusters.
Range: $[0, +\infty)$.
Sensitivity: DBI depends on k (number of clusters) and is used to compare different clustering results (e.g., choosing k via elbow plots).
Limitations: Works best for convex clusters; less reliable for complex shapes or high overlap.

2. Numeric Example

We compute DBI for a 2D dataset with **2 clusters** ($k = 2$):
Cluster 1 (C1): Points $(1, 1)$, $(1, 2)$, $(2, 1)$
Cluster 2 (C2): Points $(4, 4)$, $(5, 5)$, $(5, 4)$
Step 1: Compute Centroids
Centroid c_1 for C1:

$$c_1 = \left(\frac{1 + 1 + 2}{3}, \frac{1 + 2 + 1}{3}\right) = (1.333, 1.333)$$

Centroid c_2 for C2:

$$c_2 = \left(\frac{4+5+5}{3}, \frac{4+5+4}{3}\right) \approx (4.667, 4.333)$$

Step 2: Compute Intra-Cluster Distances (s_i)
For C1:

Distance $(1,1)$ to c_1: $\sqrt{(1-1.333)^2 + (1-1.333)^2} = \sqrt{0.222} \approx 0.471$

Distance $(1,2)$ to c_1: $\sqrt{(1-1.333)^2 + (2-1.333)^2} = \sqrt{0.556} \approx 0.746$

Distance $(2,1)$ to c_1: $\sqrt{(2-1.333)^2 + (1-1.333)^2} = \sqrt{0.556} \approx 0.746$

$s_1 = \frac{0.471+0.746+0.746}{3} \approx 0.654$

For C2:

Distance $(4,4)$ to c_2: $\sqrt{(4-4.667)^2 + (4-4.333)^2} = \sqrt{0.556} \approx 0.746$

Distance $(5,5)$ to c_2: $\sqrt{(5-4.667)^2 + (5-4.333)^2} = \sqrt{0.556} \approx 0.746$

Distance $(5,4)$ to c_2: $\sqrt{(5-4.667)^2 + (4-4.333)^2} = \sqrt{0.222} \approx 0.471$

$s_2 = \frac{0.746+0.746+0.471}{3} \approx 0.654$

Step 3: Compute Centroid Distance ($d(c_1,c_2)$)

$$d(c_1,c_2) = \sqrt{(4.667-1.333)^2 + (4.333-1.333)^2} = \sqrt{3.334^2 + 3^2} \approx \sqrt{20.115} \approx 4.485$$

Step 4: Calculate Similarity Ratios
Ratio for $i = 1, j = 2$:

$$\frac{s_1 + s_2}{d(c_1,c_2)} = \frac{0.654 + 0.654}{4.485} \approx 0.2916$$

Ratio for $i = 2, j = 1$:

$$\frac{s_2 + s_1}{d(c_2,c_1)} = \frac{0.654 + 0.654}{4.485} \approx 0.2916(symmetric)$$

Since $k = 2$, the maximum ratio for each cluster is 0.2916.
Step 5: Compute DBI

$$DBI = \frac{1}{2}(0.2916 + 0.2916) \approx 0.2916$$

Result:
DBI ≈ 0.292 (low value, indicating good clustering).
Why this makes sense: Clusters are internally compact ($s_i \approx 0.65$) and well-separated ($d(c_1,c_2) \approx 4.49$).
Reference: For $k = 2$, typical DBI values range from 0.2–0.5 in clean datasets; our result aligns with this.

Practical Notes:

Automation: Use libraries like scikit-learn (Python) for DBI calculation:

```
from sklearn.metrics import davies_bouldin_score
DBI = davies_bouldin_score(X, labels)  # X=data, labels=cluster assignments
```

Comparison: DBI values for different k (e.g., k = *3* yields higher DBI ≈ *1.086*) help identify optimal cluster counts.